Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued. . .]

Chapter 1

Superconducting instability

In this chapter the HF methods introduced in Sec. **??** are applied to the SC phase. We study a translationally invariant superconductor, which breaks the conservation of total charge. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure s-wave superconductivity.

As for previous phases, this chapter is divided in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. **??**, and implementation of HF methods. Second, search for theoretical constraints on the self-consistent solution. Third, numeric results.

1.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. **??** we anticipated that the SC phase is described by the two SSBs

$$\mathrm{U}^z(1) \overset{\mathrm{SSB}}{\to} 0$$
$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. **??**. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \tag{1.1}$$

which is identical to Eq. (**??**) for the normal phase.

Recall Tabs. **??** and **??**, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (1.2), (**??**) for the SC phase we need to account for the following contractions:

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree, Fock and Cooper contractions

We anticipate their roles in Tab. 1.1.

The SSB of $\mathrm{U}^z(1)$ allows for EVs like $\langle \hat{c}^\dagger \hat{c}^\dagger \rangle$ to be non-zero. This leads to the formation of Cooper pairs. These are composite quantum objects with a specific spin-spatial structure. Consider the generic Cooper fluctuation

$$\left\langle \hat{c}^\dagger_{i\sigma} \hat{c}^\dagger_{j\sigma'} \right\rangle \quad \text{with } i,j \text{ NN}$$

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. Now, when applying HF method to the quartic

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalization and resymmetrization
		Cooper	Anisotropic pairing in triplet channel

Table 1.1 Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. **??** and **??**.

Spatial structure	Pairing channel	RWCs in Cooper channel
Symmetric wave function	Singlet pairing	Just $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\bar{\sigma}}\rangle$
Anti-symmetric wave function	Triplet pairing	Both $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\bar{\sigma}}\rangle$ and $\langle \hat{c}^\dagger_{i\sigma}\hat{c}^\dagger_{j\sigma}\rangle$

Table 1.2 Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \to (-x, -y)$).

operators, we get the decomposition of Eq. (**??**). We can impose to the approximated HF groundstate a particular spatial structure – say, inversion symmetry or antisymmetry. This reflects in the requirement that Cooper pairs show the same spatial structure.

This gives a selection rule over Cooper pairings: if we work with inversion-symmetric structures – say, s^*-wave or d-wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarised in Tab. 1.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \tag{1.2}$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{\mathbf{k}\uparrow} \tag{1.3}$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (**??**). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

1.2 Local effects: (impossibility of) superconductivity in the HM

Let us start from the local effects. Following Tab. **??** and Eqns. (1.2), (**??**), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow} \\
&\simeq U \sum_{\mathbf{r}\in\mathcal{L}} \Big(\hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle + \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\rangle && \text{(Hartree)} \\
&\qquad + \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle + \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} - \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \Big) && \text{(Cooper)}
\end{aligned}$$

In next sections we deal with each channel separately.

1.2.1 Local Hartree channel in the SC phase: analytic derivation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. **??** for the normal phase. Substituting Eq. (1.1), we get the approximation of Eq. (**??**),

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U} + (\text{Cooper channel}) \tag{1.4}$$

thus chemical potential renormalization $\tilde{\mu} \to \mu - nU$ is present also in the SC phase.

1.2.2 Local Hartree contraction energy

As for the section above, we use the result of Sec. **??** for the normal phase. From Eq. (**??**)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U \tag{1.5}$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

1.2.3 Local Cooper channel in the SC phase: analytic derivation

We move to reciprocal space, through the transformation of Eq. (**??**). Consider the Cooper channel of Eq. (1.2). In reciprocal space we get:

$$\begin{aligned}\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \Big[&\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ &+ \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned} \tag{1.6}$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this: consider the EV of the Cooper pair

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle$$

for two momenta $\mathbf{k}_1$, $\mathbf{k}_2$. Moving to real space via an $\mathcal{F}$-transform, we get

$$\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1 \in \mathcal{L}} \sum_{\mathbf{r}_2 \in \mathcal{L}} e^{-i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)} \big\langle \hat{c}^\dagger_{\mathbf{r}_1\uparrow} \hat{c}^\dagger_{\mathbf{r}_2\downarrow} \big\rangle$$

where we used the convention of Eq. (**??**). Let now $\mathbf{r}_1 \to \mathbf{r}$ and $\mathbf{r}_2 \to \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \big\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}+\Delta\mathbf{r}\downarrow} \big\rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned}\big\langle \hat{c}^\dagger_{\mathbf{k}_1\uparrow} \hat{c}^\dagger_{\mathbf{k}_2\downarrow} \big\rangle &\overset{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}} \\ &= \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1 = \mathbf{k}_2}\end{aligned}$$

where we used the identity of Eq. (**??**). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (1.6), this proves that only pairs at null total momentum can have nonzero EVs.

Let now $w_\mathbf{k}$ be the EV of $\phi^\dagger_\mathbf{k}$, and $w^{(s)}$ its s-wave projection:

$$w_\mathbf{k} \equiv \big\langle \phi^\dagger_\mathbf{k} \big\rangle \tag{1.7}$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} w_\mathbf{k} \tag{1.8}$$

Eq. (1.6) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}^\dagger_{\mathbf{k}} \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{1.9}$$

being $E^{(\mathrm{SC})}_{\mathrm{C}/U}$ the HM Cooper contraction energy, written explicitly in Sec. 1.2.4. Let now

$$\Delta^{(s)} \equiv -U w^{(s)} \tag{1.10}$$

This is the s-wave component of the gap function. Note that, in general, $\Delta^{(s)} \in \mathbb{C}$. However, we can always choose $\Delta^{(s)}$ as purely real. This is due to the fact that the hamiltonian is gauge-invariant, then it suffices to apply the gauge map of Eq. (**??**) to the fermionic operators with

$$\eta \equiv \frac{\arg \Delta^{(s)}}{2}$$

to obtain $\Delta^{(s)} \in \mathbb{R}$. For the sake of generality, we proceed with $\Delta^{(s)} \in \mathbb{C}$. Although this choice gives no advantage now, we implement it anyway in order to derive the most general form of the equations that follow. This is useful for the generalisation to the EHM.

In the end, we reduce $\hat{H}_U$ from Eq. (1.9) to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{k} \in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] - E^{(\mathrm{SC})}_{\mathrm{C}/U} \tag{1.11}$$

This last decomposition resembles Eq. (**??**), obtained for the AF phase in the Hartree channel.

1.2.4 Local Cooper contraction energy

The Cooper contraction energy in the Cooper channel is given by:

$$\begin{aligned} E^{(\mathrm{SC})}_{\mathrm{C}/U} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}\mathbf{k}'} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle \\ &= L_x L_y \times U \left| w^{(s)} \right|^2 \end{aligned} \tag{1.12}$$

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy.

1.2.5 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. **??**. We start by giving a matrix representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matrix representation. Collecting the results of the above paragraphs, we obtain the quadratic approximation to the Hubbard hamiltonian in the SC phase,

$$\begin{aligned} \hat{H}_t + \hat{H}_U \simeq &\sum_{\mathbf{k} \in \mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} \\ &- \sum_{\mathbf{k} \in \mathrm{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}^\dagger_{\mathbf{k}} \right] + nU \times \hat{N} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \end{aligned} \tag{1.13}$$

We define the shifted bare bands:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential. Due to inversion symmetry, it holds $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. Using Fermi anticommutation rules, the kinetic operator can be written as:

$$\begin{aligned}\hat{H}_t - \tilde{\mu}\hat{N} &= \sum_{\sigma}\sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}\sigma}\hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}}\hat{n}_{-\mathbf{k}\downarrow}\right) \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow} + 1\right) \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \left(\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow}\hat{c}^{\dagger}_{-\mathbf{k}\downarrow}\right) + E_{\mathrm{a}}^{(\mathrm{SC})}\end{aligned}$$

being $E_{\mathrm{a}}^{(\mathrm{SC})}$ the "anticommutation energy"

$$E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \xi_{\mathbf{k}} \tag{1.14}$$

In the end, the hamiltonian is approximated by:

$$\begin{aligned}\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}^{\dagger}_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{k}\downarrow}\right] + E_{\mathrm{a}}^{(\mathrm{SC})} \\ - \sum_{\mathbf{q}\in\mathrm{BZ}} \left[\Delta^{(s)}\hat{\phi}_{\mathbf{q}} + \Delta^{(s)*}\hat{\phi}^{\dagger}_{\mathbf{q}}\right] + nU \times \hat{N} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right]\end{aligned}$$

with $E_{\mathrm{H}/U}^{(\mathrm{SC})}$, $E_{\mathrm{C}/U}^{(\mathrm{SC})}$ the contraction energies arising from Hartree and Cooper channels, and $E_{\mathrm{a}}^{(\mathrm{SC})}$ the energy shift resulting from anticommutation rules, given respectively in Eqns. (1.5), (1.12), and (1.14). We recognised the s-wave gap $\Delta^{(s)}$ from Eq. (1.10).

Let the Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix}$$

The approximated hamiltonian can be written in matrix form as:

$$\hat{H}_t + \hat{H}_U - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}}\sum_{\sigma} \hat{\Phi}^{\dagger}_{\mathbf{k}} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_{\mathrm{a}}^{(\mathrm{SC})} - \left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})}\right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta^*_{\mathbf{k}} \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$.

Pseudospin representation. The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}}\tau^z - \mathrm{Re}\{\Delta_{\mathbf{k}}\}\tau^x - \mathrm{Im}\{\Delta_{\mathbf{k}}\}\tau^y$$

Let now the pseudospin operator $\hat{\mathbf{s}}_{\mathbf{k}}$

$$\hat{s}^{\alpha}_{\mathbf{k}} \equiv \hat{\Phi}^{\dagger}_{\mathbf{k}}\tau^{\alpha}\hat{\Phi}_{\mathbf{k}} \quad \text{with} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}^x_{\mathbf{k}\sigma} \\ \hat{s}^y_{\mathbf{k}\sigma} \\ \hat{s}^z_{\mathbf{k}\sigma} \end{bmatrix}$$

and the pseudofield $\mathbf{b}_{\mathbf{k}}$:

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\mathrm{Re}\{\Delta_{\mathbf{k}}\} \\ -\mathrm{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}^x_{\mathbf{k}} \rangle$	$\left\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^y_{\mathbf{k}} \rangle$	$i \left\langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}} \right\rangle$
$\langle \hat{s}^z_{\mathbf{k}} \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle - 1$

Table 1.3 | List of the various expectation values (EVs) of each pseudospin component and their meaning in terms of fermionic operators.

For a purely real gap, the pseudofield has no y component. In general, different gauges rotate the pseudofield around the z axis.

As was for the AF phase, these operators satisfy a $\mathfrak{su}(2)$ algebra up to an constant, irrelevant in the context of this discussion. They satisfy the identities:

$$\hat{s}^x_{\mathbf{k}} = \hat{\phi}_{\mathbf{k}} + \hat{\phi}^{\dagger}_{\mathbf{k}} \tag{1.15}$$

$$\hat{s}^y_{\mathbf{k}} = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}^{\dagger}_{\mathbf{k}} \right) \tag{1.16}$$

$$\hat{s}^z_{\mathbf{k}} = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} - 1 \tag{1.17}$$

We report the EVs related to these identities Tab. 1.3.

We express the approximated hamiltonian in pseudospin representation as:

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \mathrm{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + nU\hat{N} + E^{(\mathrm{SC})}_{\mathrm{a}} - \left[E^{(\mathrm{SC})}_{\mathrm{H}/U} + E^{(\mathrm{SC})}_{\mathrm{C}/U} \right] \tag{1.18}$$

This operator is similar enough to the one found for the AF phase, in Eq. (**??**). It describes a set of pseudospin subject to a uniform field. As for the AF phase, we define the Bogoliubov bands $E_{\mathbf{k}}$

$$E_{\mathbf{k}} \equiv \sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2}$$

The key difference with the AF Bogoliubov bands of Eq. (**??**) is that $E_{\mathbf{k}}$ now are functions of $\tilde{\mu}$, since $\xi_{\mathbf{k}}$ depends on the chemical potential. In other words, now the Bogoliubov bands depend on the shifted bands $\xi_{\mathbf{k}}$ instead of the bare bands $\epsilon_{\mathbf{k}}$.

Moreover, let us define the angles $\theta_{\mathbf{k}}$, $\zeta_{\mathbf{k}}$ as

$$\cos 2\theta_{\mathbf{k}} = \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \qquad \sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}}$$

Note that choosing the gauge $\eta = \arg \Delta^{(s)}/2$ the imaginary part of the gap is zero, and this reflects in $2\zeta_{\mathbf{k}} = k \times 2\pi$ with $k \in \mathbb{Z}$.

Diagonalisation. For the SC phase, diagonalisation of the HF hamiltonian follows the same procedure as was for the AF phase. Indeed, the diagonalising matrix is given by

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} & -\sin\theta_{\mathbf{k}} \\ \sin\theta_{\mathbf{k}} & \cos\theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \tag{1.19}$$

and the graphic procedure is the same of Fig. **??**, with $\epsilon \to \xi$. We get:

$$d_{\mathbf{k}} = W_{\mathbf{k}} h_{\mathbf{k}} W^{\dagger}_{\mathbf{k}} \quad \text{with} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix} \tag{1.20}$$

In the tilted frame we have the expression for the Nambu spinors:

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Psi}_{\mathbf{k}} \tag{1.21}$$

Explicitly:

$$\begin{aligned}
\hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \\
&= \begin{bmatrix} \cos\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \\ \sin\theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos\theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}^{\dagger}_{-\mathbf{k}\downarrow} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}^{(+)}_{\mathbf{k}} \\ \hat{\gamma}^{(-)}_{\mathbf{k}} \end{bmatrix}
\end{aligned} \tag{1.22}$$

The new Fermionic operators describe a Fermi gas on distributed on two mirrored bands,

$$\hat{H}_t + \hat{H}_U - \tilde{\mu}\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} E_{\mathbf{k}} \left(\hat{\gamma}^{(+)\dagger}_{\mathbf{k}} \hat{\gamma}^{(+)}_{\mathbf{k}} - \hat{\gamma}^{(-)\dagger}_{\mathbf{k}} \hat{\gamma}^{(-)}_{\mathbf{k}} \right) + (\cdots)$$

with $(\cdots)$ the various energy shifts.

1.2.6 Features of the SC solution

In this section we give some details about the HF solution for the translationally-invariant SC phase. We prove that in the conventional HM, with local repulsion $\hat{H}_U$, isotropic BCS-like superconductivity is forbidden. This serves as ground to motivate the passage to the EHM.

Pseudospin EVs. The EVs of pseudospin operators in the various directions are given by:

$$\begin{aligned}
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^x \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \cos 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^y \hat{\Psi}_{\mathbf{k}} \rangle &= -\sin 2\theta_{\mathbf{k}} \sin 2\zeta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \\
\langle \hat{\Psi}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Psi}_{\mathbf{k}} \rangle &= \cos 2\theta_{\mathbf{k}} \langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle
\end{aligned}$$

In the tilted frame, the averaged z projection of pseudospin is:

$$\langle \hat{\Gamma}^{\dagger}_{\mathbf{k}} \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(E_{\mathbf{k}}; \beta, 0\right) - f\left(-E_{\mathbf{k}}; \beta, 0\right) \tag{1.23}$$

Compared with the respective relation for the AF phase, Eq. (**??**), the main difference here is that the chemical potential inside the Fermi-Dirac distribution is set to zero. This is due to the fact that $\tilde{\mu}$ is already included in the renormalised bands $E_{\mathbf{k}}$. The approximated quadratic hamiltonian describes a gas of γ fermions at null chemical potential.

Let us define the thermal factors:

$$\mathrm{Th}^{(\pm)}_{\mathbf{k}}[\beta] \equiv f\left(-E_{\mathbf{k}}; \beta, 0\right) \pm f\left(E_{\mathbf{k}}; \beta, 0\right) \tag{1.24}$$

In the end we have the EVs:

$$\langle \hat{s}^x_{\mathbf{k}} \rangle = \frac{\mathrm{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.25}$$

$$\langle \hat{s}^y_{\mathbf{k}} \rangle = \frac{\mathrm{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.26}$$

$$\langle \hat{s}^z_{\mathbf{k}} \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \tag{1.27}$$

Note that, as is expressed in Tab. 1.3, the z component of the pseudospin is related to density, being $1 + \langle \hat{s}^z_{\mathbf{k}} \rangle = \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle$.

s-wave gap self-consistency equation. Applying the HF method to the conventional HM in SC phase, we obtained just one HFP: $w^{(s)}$. Let us derive its self-consistency equation. First, let

$$\tau^{\pm} \equiv \frac{\tau^x \pm i\tau^y}{2}$$

Recall Eq. (1.8) and expand it, to obtain

$$
\begin{aligned}
w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\phi}^\dagger_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]
\end{aligned} \tag{1.28}
$$

Last line is the explicit form of the self-consistency equation for $w^{(s)}$.

Impossibility of s-wave superconductivity. Let us define the critical U,

$$
U_{\mathrm{c}}[\beta] \equiv \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \tag{1.29}
$$

We call it "critical" because its value gives a sort of boundary for convergence. Indeed, as we show in App. **??**, for the SC phase in conventional HM, we are able to optimise parametrically the mixing parameter g in order to ensure convergence, based on the ratio $U/U_{\mathrm{c}}[\beta]$.

Let us show that superconductivity is not possible in the conventional HM. Eq. (1.28) can be rewritten as:

$$
w^{(s)} = -\frac{U}{U_{\mathrm{c}}[\beta]} w^{(s)}
$$

By construction $U_{\mathrm{c}}[\beta] > 0$ (the thermal factor is negative-defined). Hence the unique possible self-consistent solution for the equation above is $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at HF level in the HM.

Density. In order to estimate density, we can use the z projection of the pseudospin operators. From Eq. (1.17) we know

$$
\hat{s}^z_{\mathbf{k}} + 1 = \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow}
$$

Then, density can be estimated as:

$$
\begin{aligned}
n &= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left(\langle \hat{s}^z_{\mathbf{k}} \rangle + 1 \right) \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left(1 - \frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \mathrm{Th}^{(-)}_{\mathbf{k}}[\beta] \right)
\end{aligned} \tag{1.30}
$$

We can now move to the EHM.

1.3 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$. As we did for the AF phase, we look for a solution with an identical mathematical structure as the one here described for the conventional HM. We define the renormalised quantities

$$
\xi_{\mathbf{k}} \to \tilde{\xi}_{\mathbf{k}} \qquad E_{\mathbf{k}} \to \tilde{E}_{\mathbf{k}} \qquad \Delta_{\mathbf{k}} \to \tilde{\Delta}_{\mathbf{k}} \qquad \mu \to \tilde{\mu}
$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. **??**. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

Let the renormalised angle:

$$\sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \qquad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tag{1.31}$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \qquad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \tag{1.32}$$

The pseudospin EVs are renormalised too,

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.33}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.34}$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \tag{1.35}$$

Let also:

$$\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) - f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{1.36}$$

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(\pm)}[\beta, \tilde{\mu}] \equiv f\left(-\tilde{E}_{\mathbf{k}}; \beta, 0\right) \pm f\left(\tilde{E}_{\mathbf{k}}; \beta, 0\right) \tag{1.37}$$

For the SC phase, the thermal factors reduce to:

$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(+)} = 1$$
$$\tilde{\mathrm{Th}}_{\mathbf{k}}^{(-)} = -\tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$$

Hence, pseudospin expectation values are:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = -\frac{\mathrm{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.38}$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = -\frac{\mathrm{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.39}$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.40}$$

These equations are an extension of Eqns. (1.38), (1.39), (1.40) for the EHM.

We can rewrite the equations for $w^{(s)}$ and n accordingly,

$$w^{(s)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \tag{1.41}$$

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{BZ}} \left[1 + \cos 2\tilde{\theta}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)\right] \tag{1.42}$$

where we used Eqns. (1.28), (1.30).

From Tabs. **??** and **??** we know none of the three channels (Hartree, Fock, Cooper) is subject to selection rules. Thus all terms of the decomposition of $\hat{H}_V$ given in Eq. (**??**) contribute. As always, we break down each.

1.4 Non-local Hartree effects: chemical potential renormalisation

We start by analysing the non-local Hartree channel,

$$\hat{H}_V \simeq -V \sum_{\langle ij\rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma} \big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\big\rangle + \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\big\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'} - \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}\big\rangle\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{j\sigma'}\big\rangle \Big) \\ + (\text{Fock channel}) + (\text{Cooper channel})$$

where we used Eq. (**??**). This discussion is based on that for the normal phase in Sec. **??**.

1.4.1 Non-local Hartree channel in the SC phase: analytic derivation

Without repeating the calculations, the non-local attraction $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/V} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E^{(\mathrm{SC})}_{\mathrm{H}/U}$$

gives the chemical potential full renormalization, identical to Eq. (**??**),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.43}$$

1.4.2 Non-local Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (**??**),

$$E^{(\mathrm{SC})}_{\mathrm{H}/V} = -L_x L_y \times 2zn^2 V$$

and together with the local counterpart of Eq. (**??**)

$$E^{(\mathrm{SC})}_{\mathrm{H}/U} = L_x L_y \times n^2 U$$

Now we move to the more interesting Fock sector.

1.5 Fock effects: bare bands renormalization

As done for the other phases, we here discuss the bands renormalization effect derived from the Fock sector of the non-local attraction. From Eq. (**??**), we have

$$\hat{H}_V \simeq (\text{Hartree channel}) \\ V \sum_{\langle ij\rangle} \sum_{\sigma,\sigma'} \Big(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\big\rangle + \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\big\rangle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma} - \big\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\big\rangle\big\langle \hat{c}^\dagger_{j\sigma'}\hat{c}_{i\sigma}\big\rangle \Big) + (\text{Cooper channel})$$

This discussion is essentially analogous the one done for the normal phase in Sec. **??**. As for previous phases, following the selection rules of Tab. **??**, only the s.s. channel contributes. Therefore we use from now on, in this section, $\sigma = \sigma'$.

1.5.1 Fock channel in the SC phase: analytic derivation

Let us define, as done for the normal phase in Eq. (**??**) and for the AF phase in Eq. (**??**),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle \tag{1.44}$$

Eq. (**??**) stands perfectly valid, translated to the SC phase

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$

$$\stackrel{\mathcal{F}}{=}\sum_{\gamma,\sigma}\left[\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}u_{\mathbf{k}\sigma}\right]\times V\sum_{\mathbf{k}'\in\mathrm{BZ}}\varphi^{(\gamma)*}_{\mathbf{k}'}\hat{c}^\dagger_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}-E^{(\mathrm{SC})}_{\mathrm{F}/V} \tag{1.45}$$

being

$$E^{(\mathrm{SC})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{k}'\sigma}\hat{c}_{\mathbf{k}'\sigma}\rangle \tag{1.46}$$

To get to this point, we employed the assumption

$$\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma'}\rangle=\delta_{\mathbf{k}=\mathbf{k}'}\delta_{\sigma=\sigma'}\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is verified by the BCS ground state of Eq. **??**. Let us define each γ-wave component as

$$u^{(\gamma)}_{\sigma}=\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi^{(\gamma)}_{\mathbf{k}}\langle\hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma}\rangle$$

These are our HFPs in the Fock channel. The SC phase does not break inversion symmetry: then, as we concluded for the other phases, p_ℓ components are null. The renormalised bands are, as in Eqns. (**??**), (**??**),

$$\tilde{\epsilon}_{\mathbf{k}}\equiv\epsilon_{\mathbf{k}}+u^{(s^*)}V(\cos k_x+\cos k_y)+u^{(d)}V(\cos k_x-\cos k_y)$$

Let us compute the EVs of $u_{\mathbf{k}\sigma}$ in the two spin sectors,

$$\begin{aligned}u_{\mathbf{k}\uparrow}&=\langle\hat{c}^\dagger_{\mathbf{k}\uparrow}\hat{c}_{\mathbf{k}\uparrow}\rangle\\&=\left\langle\hat{\Phi}^\dagger_{\mathbf{k}}\frac{\mathbb{I}+\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=\frac{1}{2}\left[\left\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle+\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]-\cos(2\tilde{\theta}_{\mathbf{k}})\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=\frac{1}{2}\left[1-\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned} \tag{1.47}$$

having we used Eq. (**??**) and following. For the spin $\downarrow$ EV,

$$\begin{aligned}u_{-\mathbf{k}\downarrow}&=\langle\hat{c}^\dagger_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}\rangle\\&=1-\langle\hat{c}_{-\mathbf{k}\downarrow}\hat{c}^\dagger_{-\mathbf{k}\downarrow}\rangle\\&=1-\left\langle\hat{\Phi}^\dagger_{\mathbf{k}}\frac{\mathbb{I}-\tau^z}{2}\hat{\Phi}_{\mathbf{k}}\right\rangle\\&=1-\frac{1}{2}\left[\left\langle\hat{\Gamma}^\dagger_{\mathbf{k}}\hat{\Gamma}_{\mathbf{k}}\right\rangle-\langle\hat{s}^z_{\mathbf{k}}\rangle\right]\\&=1-\frac{1}{2}\left[\tilde{\mathrm{Th}}^{(+)}_{\mathbf{k}}[\beta]+\cos 2\tilde{\theta}_{\mathbf{k}}\tilde{\mathrm{Th}}^{(-)}_{\mathbf{k}}[\beta]\right]\\&=1-\frac{1}{2}\left[1+\cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]\end{aligned} \tag{1.48}$$

From the above results, note first that

$$u_{\mathbf{k}\uparrow}=u_{-\mathbf{k}\downarrow}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x + \cos k_y)\left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. **??**
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}(\cos k_x - \cos k_y)\left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]$	Fig. **??**

Table 1.4 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. Note also that Eqns. (1.47), (1.48) combine to give the expression for density of Eq. (1.42).

If we take the $\tilde{\theta}_{\mathbf{k}} \to 0$ limit, we must retrieve the Fermi-Dirac distribution. Indeed, this limit coincides with $U \to 0$. Explicitly:

$$\begin{aligned}
\lim_{\tilde{\theta}_{\mathbf{k}}\to 0} u_{\mathbf{k}\sigma} &= \frac{1}{2}\left[1 + \tanh\left(\frac{\beta\tilde{\xi}_{\mathbf{k}}}{2}\right)\right] \\
&= \frac{1}{2}\left[\frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}} - \frac{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} - e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta\tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta\tilde{\xi}_{\mathbf{k}}/2}}\right] \\
&= f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

We can then derive the $u_{\mathbf{k}}$-related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi_{\mathbf{k}}^{(\gamma)}\langle\hat{n}_{\mathbf{k}\uparrow}\rangle \\
&= \frac{1}{2L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\varphi_{\mathbf{k}}^{(\gamma)}\left[1 - \cos(2\tilde{\theta}_{\mathbf{k}})\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\right]
\end{aligned} \tag{1.49}$$

reported explicitly for the only two inversion symmetric structures in Tab. 1.4.

1.5.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (**??**) holds identically

$$E_{\mathrm{F}/V}^{(\mathrm{SC})} = L_xL_y \times V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (1.47), (1.48).

1.6 Non-local Cooper effects: anisotropic pairing

In this section we deal with the Cooper part of Eq. (**??**),

$$\begin{aligned}
\hat{H}_V =\simeq &\ (\text{Hartree channel}) + (\text{Fock channel}) \\
&- V\sum_{\langle ij\rangle}\sum_{\sigma,\sigma'}\left(\hat{c}_{i\sigma}^\dagger\hat{c}_{j\sigma'}^\dagger\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle + \langle\hat{c}_{i\sigma}^\dagger\hat{c}_{j\sigma'}^\dagger\rangle\hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \langle\hat{c}_{i\sigma}^\dagger\hat{c}_{j\sigma'}^\dagger\rangle\langle\hat{c}_{j\sigma'}\hat{c}_{i\sigma}\rangle\right)
\end{aligned}$$

We know from Eq. (**??**) that the non-local attraction can be separated in o.s. and s.s. channels. As we stated in Sec. 1.1, we need to distinguish between singlet and triplet Cooper pairing.

1.6.1 Non-local Cooper o.s. channel in the SC phase: analytic derivation

The o.s. Cooper sector contributes both to singlet and triplet pairing. Note that in singlet pairing channel the spatial wavefunction must be inversion-symmetric. On the contrary, in triplet pairing channel it needs to be inversion-antisymmetric.

We move to reciprocal space. From Eq. (**??**), we get

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} \simeq\ & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \\ & + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned}$$

Identical considerations as for the local repulsion hold, and uniquely the $\mathbf{K} = \mathbf{0}$ term gives a non-zero contribution, in general. Therefore:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} \simeq\ & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \times \Big[\big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \\ & + \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle - \big\langle \hat{c}^\dagger_{\mathbf{k}\uparrow} \hat{c}^\dagger_{-\mathbf{k}\downarrow} \big\rangle \big\langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \big\rangle \Big]\end{aligned}$$

Then, using the harmonics decomposition of Eq. (**??**), we end up with:

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w^*_{\mathbf{k}} \hat{\phi}^\dagger_{\mathbf{k}'} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \tag{1.50}$$

FROM HERE...

In the above equation the contraction energy is given by

$$E^{(\text{SC})}_{\text{C}/V/\text{o.s.}} \equiv -\frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k},\mathbf{k}'} \varphi^{(\gamma)}_{\mathbf{k}} \varphi^{(\gamma)*}_{\mathbf{k}'} w_{\mathbf{k}} w^*_{\mathbf{k}'} \tag{1.51}$$

Let then the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} w_{\mathbf{k}}$$

Eq. (1.50) becomes

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - V \sum_{\gamma} \sum_{\mathbf{k}\in\text{BZ}} \left[w^{(\gamma)} \varphi^{(\gamma)*}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E^{(\text{SC})}_{\text{C}/V/\text{o.s.}}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

For the γ-wave component of the superconducting order parameter, the following chain, similar to Eq. (1.28), holds:

$$\begin{aligned} w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \big\langle \hat{\phi}^\dagger_{\mathbf{k}} \big\rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \big\langle \hat{\Phi}^\dagger_{\mathbf{k}} \tau^+ \hat{\Phi}_{\mathbf{k}} \big\rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\langle \hat{s}^x_{\mathbf{k}} \rangle + i \langle \hat{s}^y_{\mathbf{k}} \rangle}{2} \\ &= \frac{1}{2 L_x L_y} \sum_{\mathbf{k}\in\text{BZ}} \varphi^{(\gamma)}_{\mathbf{k}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \text{Th}^{(-)}_{\mathbf{k}}[\beta] \end{aligned} \tag{1.52}$$

Symmetry	Self-consistency equation	Graph
s-wave	$w^{(s)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
s^*-wave	$w^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_x-wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_y-wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
$d_{x^2-y^2}$-wave	$w^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??

Table 1.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$.

where $\tau^+ = (\tau^x + i\tau^y)/2$. We used Eqns. (1.33), (1.34) and (1.36) as well as the thermal factor defined in Eq. (1.24). The derivation above holds for all symmetries, (including s-wave). Explicitly, five self-consistency equations are then derived and listed in Tab. 1.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter uniquely in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

1.6.2 Non-local Cooper o.s. contraction energy

The Cooper contraction energy is immediately derived from Eq. (1.51) by recognizing the harmonics of $w_{\mathbf{k}}$,

$$E^{(\mathrm{SC})}_{\mathrm{C}/V/\mathrm{o.s.}} = -L_x L_y \times V \sum_{\gamma} \left|w^{(\gamma)}\right|^2$$

and is completed by its local counterpart

$$E^{(\mathrm{SC})}_{\mathrm{C}/U} = L_x L_y \times U \left|w^{(s)}\right|^2$$

discussed for the conventional HM in Eq. (??).

1.6.3 Non-local Cooper s.s. channel in the SC phase: analytical considerations

Consider Eq. (??) in the Cooper channel,

$$\begin{aligned}\hat{H}^{(\mathrm{s.s.})}_V \simeq\ & (\text{Hartree channel}) + (\text{Fock channel}) \\ & - \frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \Big[\langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\ & + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle - \langle \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \Big]\end{aligned}$$

In principle, this pairing channel is legitimate and gives rise to "polarized" (same spin) Cooper class RWCs. This kind of spin pairing couples with antisymmetric spatial structures, working in the triplet pairing channel. However, we can safely ignore and set to zero these EVs, as is proven [17, 18]. The reason is simply that the superconducting phase preserves SU(2) spin symmetry, thus all three triplet pairing subspaces

$$z \text{ spin projection} \in \{-1, 0, 1\}$$

are degenerate and give rise to an identical solution. We can safely use the most convenient to us, the one with null spin projection, which can be mapped on the HM model of superconductivity discussed in Sec. **??**.

Anyhow, for the sake of completeness, the s.s. pairing channel is explicitly discussed in Sec. [Missing]. As is seen there, following the sketch of [18], the resulting triplet of HFPs can be safely reduced to the simplified set of o.s. triplet HFPs discussed in previous sections.

1.7 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_{\mathrm{c}}^{(\mathrm{SC})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{SC})} + E_{\mathrm{C}/U}^{(\mathrm{SC})} + E_{\mathrm{H}/V}^{(\mathrm{SC})} + E_{\mathrm{F}/V}^{(\mathrm{SC})} + E_{\mathrm{C}/V/\mathrm{o.s.}}^{(\mathrm{SC})}\right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^{\dagger}\hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}\hat{c}_{-\mathbf{k}\downarrow}^{\dagger}\right] - \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\Delta}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}} + \tilde{\Delta}_{\mathbf{k}}^{*}\hat{\phi}_{\mathbf{k}}^{\dagger}\right] + E_{\mathrm{a}}^{(\mathrm{SC})} + E_{\mathrm{c}}^{(\mathrm{SC})}$$

having we included the "anticommutation energy" also defined for the conventional HM in Eq. (**??**),

$$E_{\mathrm{a}}^{(\mathrm{SC})} \equiv \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V \sum_{\gamma\in\{s^*,d\}} u^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} - \tilde{\mu} \tag{1.53}$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -Uw^{(s)} + V\sum_{\gamma} w^{(\gamma)}\varphi_{\mathbf{k}}^{(\gamma)*} \tag{1.54}$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -Uw^{(s)} \qquad \tilde{\Delta}^{(\gamma)} = Vw^{(\gamma)} \quad \text{being} \quad \gamma \in \{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)}(\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)}(\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2}\left[\tilde{\Delta}^{(p_x)}\sin k_x + \tilde{\Delta}^{(p_y)}\sin k_y\right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalized ground-state.

1.7.1 The relative phase in the zero temperature half-filled ground state

The conventional BCS ground state is widely known and derived for HM superconductivity in Sec. **??**. Jumping to Eq. (**??**) and extending it to the present context of the EHM, we get

$$|\Psi\rangle = \bigotimes_{\mathbf{k}\in\mathrm{BZ}} \left[\cos\tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}}\sin\tilde{\theta}_{\mathbf{k}}\hat{\phi}_{\mathbf{k}}^{\dagger}\right]|\Omega_{\mathbf{k}}\rangle$$

a BCS state described by the RMPs. Within this short section, we just illustrate the most interesting features of this state. First, the phase factor angle is the gap "tilt" in the complex plane. This is evident by the facts

$$\begin{aligned}\langle\Psi|\hat{s}^x_{\mathbf{k}}|\Psi\rangle &= \langle\Psi|\hat{\phi}_{\mathbf{k}}+\hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle\\ &= \sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{2i\tilde{\zeta}_{\mathbf{k}}}+e^{-2i\tilde{\zeta}_{\mathbf{k}}}\right)\\ &= \sin(2\tilde{\theta}_{\mathbf{k}})\cos(2\tilde{\zeta}_{\mathbf{k}})\end{aligned}$$

$$\begin{aligned}\langle\Psi|\hat{s}^y_{\mathbf{k}}|\Psi\rangle &= i\,\langle\Psi|\hat{\phi}_{\mathbf{k}}-\hat{\phi}^\dagger_{\mathbf{k}}|\Psi\rangle\\ &= i\sin\tilde{\theta}_{\mathbf{k}}\cos\tilde{\theta}_{\mathbf{k}}\left(e^{-2i\tilde{\zeta}_{\mathbf{k}}}-e^{2i\tilde{\zeta}_{\mathbf{k}}}\right)\\ &= \sin(2\tilde{\theta}_{\mathbf{k}})\sin(2\tilde{\zeta}_{\mathbf{k}})\end{aligned}$$

results that are coherent with Eqns. (1.33) as well as (1.34) and allow us to identify the phase with the pseudofield xy angle.

1.8 SC free energy density

Free energy density of the superconducting phase is derived as for the normal phase in Sec. ??. As for the AF phase of Sec. ??, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (??), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$\begin{aligned}s^{(\pm)} \equiv -\frac{k_B}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\Big[&\left(1-f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\right)\ln\left(1-f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\right)\\ &+f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\ln f(\pm\tilde{E}_{\mathbf{k}};\beta,0)\Big]\end{aligned} \tag{1.55}$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^x+1}+\frac{1}{e^{-x}+1}=1$$

it's immediate to conclude that

$$s^{(-)}=s^{(+)} \qquad \Longrightarrow \qquad s\equiv s^{(-)}+s^{(+)}=2s^{(+)}$$

being s the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:
$$\begin{aligned}e^{(\mathrm{SC})}_{\mathrm{g}} &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\mathrm{Th}^{(-)}_{\mathbf{k}}[\beta]\\ &= -\frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{E}_{\mathbf{k}}\tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)\end{aligned}$$

2. Anticommutation energy:
$$\begin{aligned}e^{(\mathrm{SC})}_{\mathrm{a}} &\equiv \frac{E^{(\mathrm{SC})}_{\mathrm{a}}}{L_xL_y}\\ &= \frac{1}{L_xL_y}\sum_{\mathbf{k}\in\mathrm{BZ}}\tilde{\xi}_{\mathbf{k}}\end{aligned}$$

3. Contraction energy:
$$e^{(\mathrm{SC})}_{\mathrm{c}} = -U\left[n^2+\left|w^{(s)}\right|^2\right]+V\left[2zn^2-\sum_{\gamma}\left|u^{(\gamma)}\right|^2+\sum_{\gamma}\left|w^{(\gamma)}\right|^2\right]$$

4. Entropic energy

$$e_s^{(\mathrm{SC})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}};\beta,0)$$

5. Chemical potential correction

$$e_n^{(\mathrm{SC})} = \mu n$$

Compose everything

$$f_{\mathrm{MFT}}^{(\mathrm{SC})} \equiv e_{\mathrm{g}}^{(\mathrm{SC})} + e_{\mathrm{a}}^{(\mathrm{SC})} + e_{\mathrm{c}}^{(\mathrm{SC})} + e_s^{(\mathrm{SC})} + e_n^{(\mathrm{SC})}$$

and use that:

$$\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x + 1} = 1$$

In the end we get:

$$f_{\mathrm{MFT}}^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right)$$
$$- U\left[n^2 + \left|w^{(s)}\right|^2\right] + V\left[2zn^2 - \sum_\gamma \left|u^{(\gamma)}\right|^2 + \sum_\gamma \left|w^{(\gamma)}\right|^2\right] + \mu$$

As was for the normal and AF phases, we recognize the chemical potential shift of Eq. (1.43),

$$f_{\mathrm{MFT}}^{(\mathrm{SC})} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}}\right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{E}_{\mathbf{k}};\beta,0)\right)$$
$$- U\left|w^{(s)}\right|^2 - V\left[\sum_\gamma \left|u^{(\gamma)}\right|^2 - \sum_\gamma \left|w^{(\gamma)}\right|^2\right] + \tilde{\mu} n \quad (1.56)$$

1.8.1 The gap energy

Similarly to Sec. ??, let us prove the following equation about the gap energy:

$$-U\left|w^{(s)}\right|^2 + V\sum_\gamma \left|w^{(\gamma)}\right|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) \quad (1.57)$$

Expand the right-hand side through Eq. (1.54),

$$\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\left|\tilde{\Delta}_{\mathbf{k}}\right|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right) = \frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[-Uw^{(s)*} + V\sum_\gamma w^{(\gamma)*}\varphi_{\mathbf{k}}^{(\gamma)}\right] \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)$$
$$= -Uw^{(s)*} \times \overbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}}$$
$$+ V\sum_\gamma w^{(\gamma)*} \times \underbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta\tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}$$

having we used the self-consistency equations of Tab. 1.5. This proves Eq.(1.57), which is pretty similar to Eq. (??) with one important exception. Since the right-hand side of Eq.(1.57) is positive by definition, for $V > 0$ we have the constraint

$$\sum_\gamma \left|w^{(\gamma)}\right|^2 \geq M \quad \text{being} \quad M^{(\mathrm{SC})} \equiv \left|w^{(s)}\right|^2 \frac{U}{V}$$

Now, as we state in Sec. ??, for the pure HM we have identically $w^{(s)} = 0$. The pure HM cannot express BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U \ll V$, even small s-wave gaps impose larger $\gamma \neq s$ components;
- The s-wave gap $w^{(s)}$ cannot rise without the contributions of more complex gap topologies, which translates in (with an uncivil abuse of notation)

  $$\bigotimes_{\gamma \neq s} \lim_{w^{(\gamma)} \to 0} w^{(s)} = 0$$

  The unconventional limit notation indicates a limit being taken contemporarily on all symmetries except the s-wave one.

Of course, Eq. (1.57) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

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